

21 The molecular formula of compound **X** is $\text{C}_5\text{H}_{12}\text{O}$. The following shows the reactions compound **X** undergoes:

- reacts with alkaline aqueous iodine
- can be dehydrated to form two alkenes only

What could be the identity of compound **X**?

- A** $(\text{CH}_3)_2\text{CHCH}(\text{OH})\text{CH}_3$
- B** $(\text{CH}_3)_2\text{C}(\text{OH})\text{CH}_2\text{CH}_3$
- C** $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{OH}$
- D** $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$